

ARDUINO IOT CLOUD

University of Alexandria, Faculty of Engineering,
Computer and Systems Engineering Department.
CSE343 Embedded Systems Spring 2026

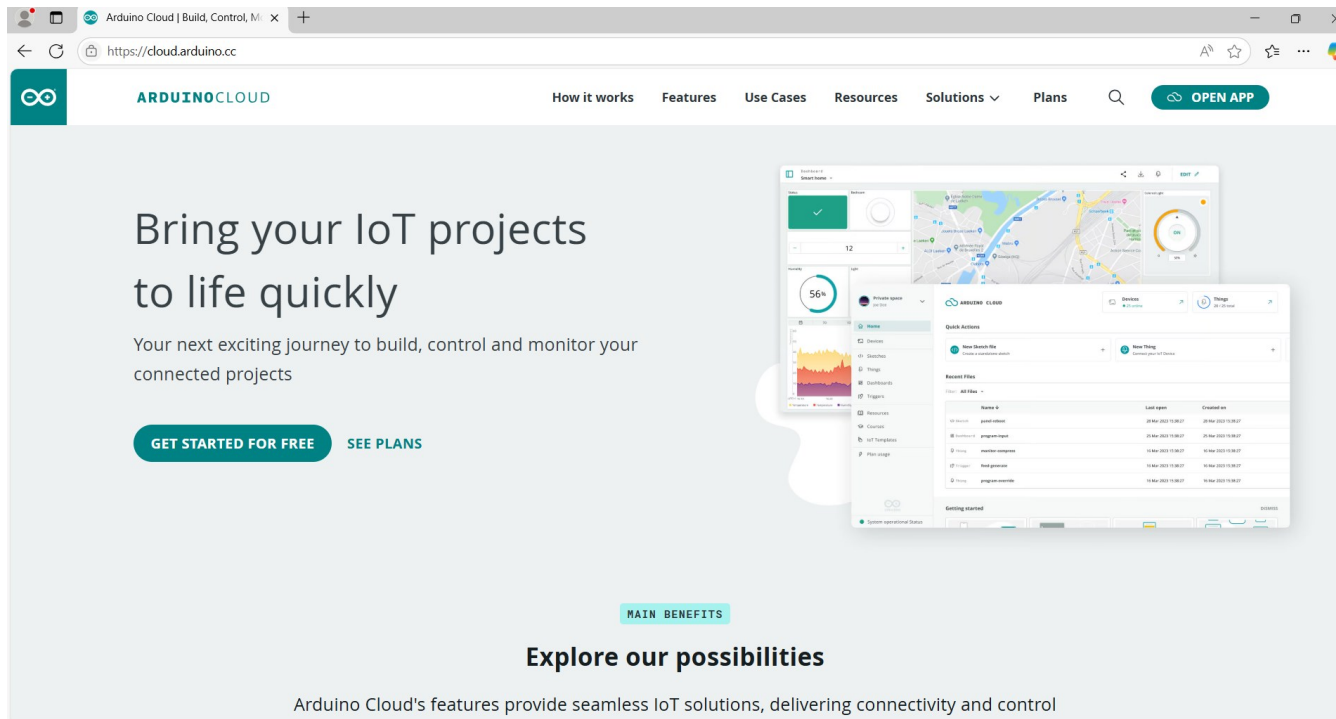
IoT

- IoT → Evolution of Embedded Systems
 - Sensors read data from surrounding environment continually.
 - Don't use this data until microcontroller asks for it.
 - Data from embedded systems are used to get information.
 - Data to information transformation.
 - Currently embedded systems are disconnected from global environments.
 - Connect the sensors through the internet.
 - Producer-Consumer concept
 - Producer → Sensor
 - Consumer → Many machines called clients.
 - Broker → Server

Arduino IoT Cloud

- IoT Platform
 - Like AWS, Azure, ..etc.
- Data Monitoring and Visualization
- Dashboard Sharing
- Mobile App Control
- Various Features

<https://cloud.arduino.cc/>

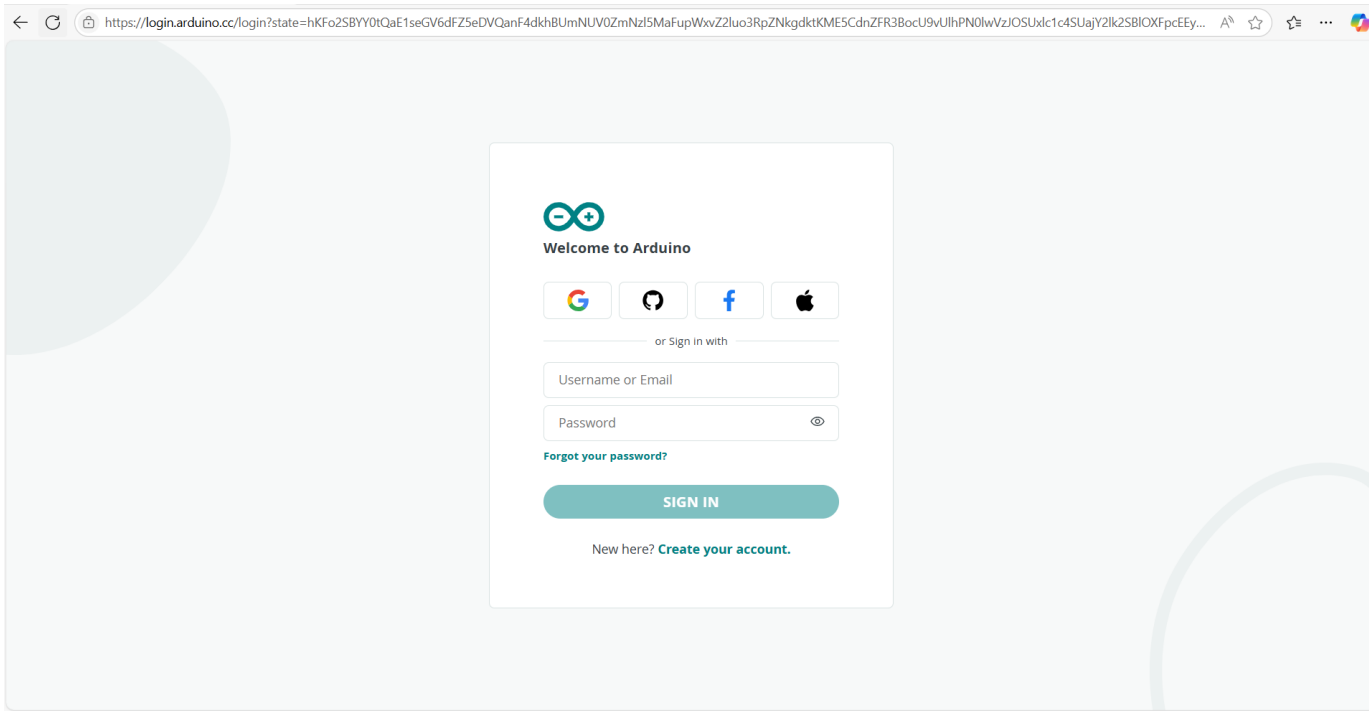


The screenshot shows the Arduino Cloud website interface. At the top, there's a navigation bar with the Arduino Cloud logo, a search icon, and an 'OPEN APP' button. The main heading reads 'Bring your IoT projects to life quickly', followed by the subtext 'Your next exciting journey to build, control and monitor your connected projects'. Below this are two buttons: 'GET STARTED FOR FREE' and 'SEE PLANS'. To the right, a preview of the Arduino Cloud dashboard is shown, featuring a map, a gauge showing 56%, and a table of recent data.

MAIN BENEFITS

Explore our possibilities

Arduino Cloud's features provide seamless IoT solutions, delivering connectivity and control



csedmicroproce...

csed.microprocessor...

Home

Sketches

Devices

Things

Dashboards

Triggers

Templates

Resources

Courses

Integrations

Plan Usage

Arduino Cloud

Ready to go beyond this free

All Systems Operational

←

↺

https://app.arduino.cc/things

☆

☆

...

∞

Create a Thing

A Thing is a virtual entity that lets you link your physical device to the Cloud: it includes variables, sketch and metadata

Associate Device and Network

Select the Device you want to use and enter your network credentials, so you can send and receive data remotely

Start creating!

Easily add Cloud Variables that will be automatically included in the sketch, and that are used to exchange data with the Cloud

+ CREATE THING

Setup Device



RECOMMENDED

AUTOMATIC Auto-sketch generation, online editing, easy upload



Arduino board ⓘ

⌘ Arduino language (C++)



Third party device ⓘ

⌘ Arduino language (C++)

MANUAL Manual programming and upload, offline editing



Any Device ⓘ

⌘ Python, MicroPython, JavaScript (NodeJS)



Setup Device



Select device type

Please select the device type and model you want to configure

☐

ESP8266

☒

ESP32

☐

LoRaWAN

ESP32-WROOM-DA Module



CONTINUE



Setup Device



Give your device a name

Name your device so you will be able to recognize it.

Device Name

as-test



NEXT

Make your device IoT-ready

To use this board you will need a Device ID and a Secret Key, please copy and save them or download the PDF. Also, keep in mind that this device authentication has a lower security level compared to other Arduino devices.

Device ID

fd085f34-a066-4f50-b348-137e83b289b1

Secret Key

enkDu2!8EZOCYQeRd7u59u46?

[DOWNLOAD PDF](#)**Secret key cannot be recovered**

Please keep it safe, if you lose it you will have to delete and setup your device again.



I saved my Device ID and Secret Key

[CONTINUE](#)

Setup device



Congratulations! You are all set

Your ESP32-WROOM-DA Module as-test has been successfully set up. You can now connect your device to sensors, actuators and other inputs or outputs!

DONE

Devices > as-test

ESP32-WROOM-DA Module

● OFFLINE

Last Activity

Apr 8, 2025, 12:11:00 AM

Added

Apr 8, 2025, 12:09:33 AM

ID

fd085f34-a066-4f50-b348-137e83b289b1

FQBN

esp32:esp32:esp32da

Associated Thing

ESP

No Thing associated with as-test

Device Status History

View the history of your device's status over selected time ranges.

15 days

7 days

1 days

1 hour

Live

No history for this device.

OTA Updates

View the status and details of your over-the-air (OTA) updates.

Start Date

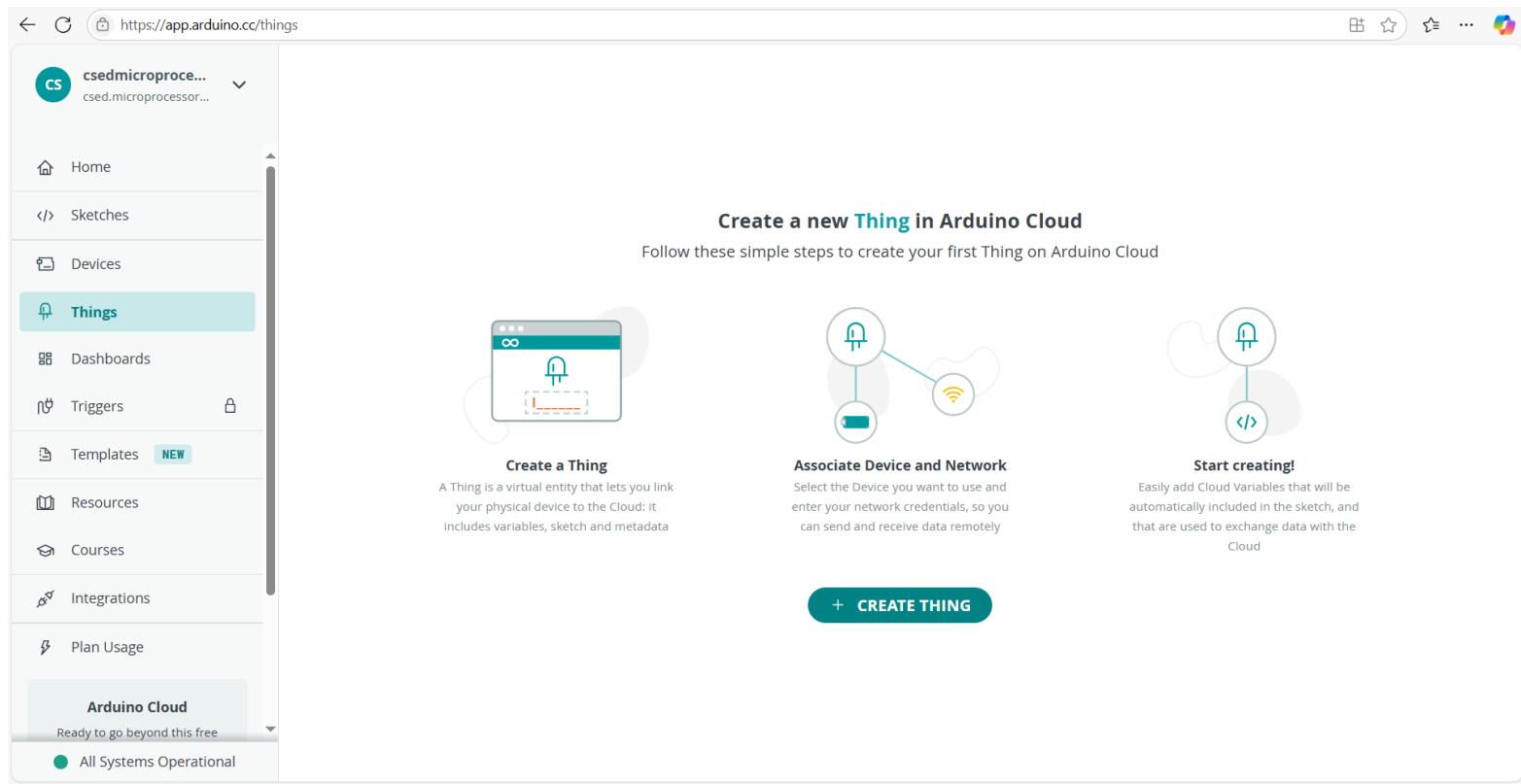
Duration

Details

Attempts

Outcome

No OTA available for this device.



cs

csedmicroproce...

csed.microprocessor...

Home

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Ready to go beyond this free

All Systems Operational

Things

Search and filter Things

View

+ THING

Thing Name ↑	Device	Sketch	Last modified	Creation date	
Untitled	-	Untitled_apr07a	Apr 8, 2025, 12:12 AM	Apr 8, 2025, 12:12 AM	Open in new tab Rename Duplicate Delete

Type here to search

18°C

سما صافية

ENG

12:15 AM

4/8/2025

Rename thing



Rename thing

Name *

as-test-thing

CONTINUE

←

↺

https://app.arduino.cc/things/8eb1e64a-10bd-4e4b-810d-aef62a08d8fd/setup

☆

☆

...

Things > Untitled ▾

⊖

⊕

⚙️ Setup

</> Sketch

📄 Metadata

Cloud Variables

ADD

Variables are what you can monitor or control to make your Thing function. For example a temperature or a smart lamp. Once created, you can use them in your Sketch.

Associated Device

Select the device you want to use or configure a new one.

🔌

Select Device

Network

Enter your network credentials to connect your device.

🌐

Configure

Smart Home integration

Configure your Thing to work with Amazon Alexa or Google Home

🔌

Configure

Add variable

×

Name

pot_value



Sync with other Things

i

Floating Point Number

eg. 1.55

▼

Declaration

`float pot_value ;`

i

Variable Permission

i

☐

Read & Write

☒

Read Only

CANCEL

ADD VARIABLE

Add variable

×

Name

pot_threshold



Sync with other Things 

Floating Point Number

eg. 1.55

▼

Declaration

`float pot_threshold ;`



Variable Permission 

☒ Read & Write

☐ Read Only

CANCEL

ADD VARIABLE



Things > as-test-thing ▾



 Setup

 Sketch ●

 Metadata

Cloud Variables

ADD

Name ▾	Last Value	Last Update
☐ pot_threshold float pot_threshold;	-	
☐ pot_value float pot_value;	-	

Associated Device

Select the device you want to use or configure a new one.



Select Device

Network

Enter your network credentials to connect your device.



Configure

Smart Home integration

Configure your Thing to work with Amazon Alexa or Google Home



Configure



Associate device



A device is ready to be associated:



as-test

ESP32-WROOM-DA
Module

ASSOCIATE

SET UP NEW DEVICE

Things > as-test-thing

Setup

</> Sketch

Metadata

Cloud Variables

ADD

	Name ↓	Last Value	Last Update
☐	pot_threshold float pot_threshold;	-	
☐	pot_value float pot_value;	-	

Change

Detach

Network

Enter your network credentials to connect your device.

Configure

Smart Home integration

Configure your Thing to work with Amazon Alexa or Google Home

Configure

Data forwarding (Webhook) ⓘ

Send data from your Thing to external services

Configure network



Enter your network credentials to allow your device to connect to the Cloud.



IMPORTANT: Remember to go to the "**Sketch**" tab and upload the sketch to load the credentials on the board.

SAVE

Configure network



Enter your network credentials to allow your device to connect to the Cloud.

Wi-Fi Name *

Your wifi name

Password

Your wifi password



Secret Key *

lenkDu2!8EZOCYQeRd7u59u46?



IMPORTANT: Remember to go to the "Sketch" tab and upload the sketch to load the credentials on the board.

SAVE

Untitled_apr07a.inoReadMe.adocthingProperties.hSketch Secrets+

```
// Code generated by Arduino IoT Cloud, DO NOT EDIT.

#include <ArduinoIoTCloud.h>
#include <Arduino_ConnectionHandler.h>

const char DEVICE_LOGIN_NAME[] = "fd085f34-a066-4f50-b348-137e83b289b1";

const char SSID[] = SECRET_SSID; // Network SSID (name)
const char PASS[] = SECRET_OPTIONAL_PASS; // Network password (use for WPA, or use as key for WEP)
const char DEVICE_KEY[] = SECRET_DEVICE_KEY; // Secret device password

void onPotThresholdChange();

float pot_threshold;
float pot_value;

void initProperties(){

  ArduinoCloud.setBoardId(DEVICE_LOGIN_NAME);
  ArduinoCloud.setSecretDeviceKey(DEVICE_KEY);
  ArduinoCloud.addProperty(pot_threshold, READWRITE, ON_CHANGE, onPotThresholdChange);
  ArduinoCloud.addProperty(pot_value, READ, ON_CHANGE, NULL);

}

WiFiConnectionHandler ArduinoIoTPreferredConnection(SSID, PASS);
```

sole



as-test - ESP32-WROOM-DA Module



Serial Monitor



Untitled_apr07a.ino

ReadMe.adoc

thingProperties.h

Sketch Secrets



Hide your sensitive data

You can use Secrets to hide sensitive data (passwords, API keys, etc.) so that they are not publicly displayed when you share your Sketches. A secret consists of a Key, the part that remains visible when the Sketch is shared, and a Value, visible only to the user who created the Sketch.

Key

SECRET_DEVICE_KEY



Value

.....



Key

SECRET_OPTIONAL_PASS



Value

.....



Key

SECRET_SSID



Value

.....



ADD NEW





as-test - ESP32-WROOM-DA Module



Serial Monitor



Untitled_apr07a.ino

ReadMe.adoc

thingProperties.h

Sketch Secrets



```
31  /*
32     The following function allows you to obtain more information
33     related to the state of network and IoT Cloud connection and errors
34     the higher number the more granular information you'll get.
35     The default is 0 (only errors).
36     Maximum is 4
37  */
38  setDebugMessageLevel(2);
39  ArduinoCloud.printDebugInfo();
40  }
41
42  void loop() {
43    ArduinoCloud.update();
44    // Your code here
45
46
47  }
48
49
50  /*
51  Since PotThreshold is READ_WRITE variable, onPotThresholdChange() is
52  executed every time a new value is received from IoT Cloud.
53  */
54  void onPotThresholdChange() {
55    // Add your code here to act upon PotThreshold change
56  }
57
```



Things > as-test-thing

Setup Sketch Metadata

as-test - ESP32-WROOM-DA Module

Serial Monitor

Untitled_apr07a.ino

ReadMe.adoc

thingProperties.h

Sketch Secrets

```
1  /*
2   Sketch generated by the Arduino IoT Cloud Thing "Untitled"
3   https://create.arduino.cc/cloud/things/8eb1e64a-10bd-4e4b-810d-aef62a08d8fd
4
5   Arduino IoT Cloud Variables description
6
7   The following variables are automatically generated and updated when changes are made to the Thing
8
9   float pot_threshold;
10  float pot_value;
11
12  Variables which are marked as READ/WRITE in the Cloud Thing will also have functions
13  which are called when their values are changed from the Dashboard.
14  These functions are generated with the Thing and added at the end of this sketch.
15  */
16
17  #include "thingProperties.h"
```

Console

←

→

⌵

⬆

This PC > Documents > Arduino

⌵

↺

🔍 Search Arduino

★ Quick access

📁 Desktop

📁 Downloads

📁 Documents

📁 Pictures

📁 [4]-Open Vocabulary Object De

📁 [6]-Aerial Object Detection

📁 Breadth of Knowledge

📁 final

📁 OneDrive

📁 This PC

📁 3D Objects

📁 Desktop

📁 Documents

📁 Downloads

📁 Music

📁 Pictures

📁 Videos

📁 OS (C:)

📁 DATA (D:)

📁 Network

Name	Date modified	Type	Size
riavis	4/22/2022 8:17 PM	File folder	
scanwifiesp32	5/17/2022 11:05 AM	File folder	
simple_esp32_sketch	5/8/2023 8:56 PM	File folder	
sket_arduino_08_05	5/10/2022 8:23 PM	File folder	
sketch_aug28a	8/28/2022 12:15 PM	File folder	
sketch_buzzer_raw	4/20/2022 1:08 PM	File folder	
sketch_esp32_hello_world	11/1/2023 12:25 AM	File folder	
sketch_gesture_crossEntropy	1/19/2025 11:35 PM	File folder	
sketch_gesture_mae	1/21/2025 11:08 PM	File folder	
sketch_gesture_modified	10/26/2024 3:16 AM	File folder	
sketch_gesture_mse	1/21/2025 10:50 PM	File folder	
sketch_mar13a	3/13/2022 4:18 PM	File folder	
sketch_mar29a	3/29/2023 4:52 PM	File folder	
sketch_may02a	5/2/2023 10:16 AM	File folder	
sketch_may13a	5/13/2022 1:45 PM	File folder	
sketch_nov01a	11/1/2023 7:17 PM	File folder	
sketch_oximeter	4/20/2022 11:49 AM	File folder	
sketch_rgb	4/20/2022 11:28 AM	File folder	
sketch_servo_basic	4/20/2022 12:15 PM	File folder	
teapotdemo	3/25/2022 12:18 AM	File folder	
Untitled_apr07a	4/8/2025 12:30 AM	File folder	
WarningSystem	11/4/2022 11:06 PM	File folder	
WiFiClientSecure	5/17/2022 9:20 PM	File folder	
WiFiMulti	5/18/2022 1:49 PM	File folder	
WiFiScan	5/17/2022 9:34 PM	File folder	
wifitestesp32	5/17/2022 1:06 PM	File folder	
as-test_apr01a.zip	4/1/2025 11:29 PM	WinRAR ZIP archive	5 KB
Untitled_apr07a.zip	4/8/2025 12:28 AM	WinRAR ZIP archive	5 KB

34 items 1 item selected

The screenshot shows the Arduino IDE 2.1.0 interface. The main editor displays a simple C++ sketch with the following code:

```
// This is a task.  
  
return or exit.
```

The Boards Manager window is open, showing a list of boards. The 'Boards Manager...' option is selected, opening a sub-menu with the following options:

- Boards Manager... (Ctrl+Shift+B)
- Arduino AVR Boards
- Arduino Mbed OS Nano Boards
- Arduino SAMD Boards (32-bits ARM Cortex-M0+)
- esp32

The Windows taskbar is visible at the bottom, showing the Start button, search bar, and various application icons. The system tray shows the date and time: 12:36 AM, 4/8/2025.

BOARDS MANAGER

esp32

Type: All

Arduino ESP32 Boards by Arduino

Boards included in this package:
Arduino Nano ESP32
More Info

2.0.18-

INSTALL

esp32 by Espressif Systems

3.2.0 installed

Boards included in this package:
Pycom GPy, ATD1.47-S3, YelloByte YB-ESP32-S3-AMP (Rev.3), Heltec Wireles...
More Info

3.2.0

REMOVE

TaskStatus.ino

```
72
73 void TaskBlink(void *pvParameters) // This is a task.
74 {
75     (void) pvParameters;
76     pinMode(LED_BUILTIN, OUTPUT);
77     for (;;) // A Task shall never return or exit.
78     {
79         //Serial.println(11);
80         digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
81         vTaskDelay( 1000 / portTICK_PERIOD_MS ); // wait for one second
82         digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW
83         vTaskDelay( 1000 / portTICK_PERIOD_MS ); // wait for one second
84     }
85 }
86
```

Output

⚠ Connection lost. Cloud sketch actions and updates won't be available. ×

ℹ Updates are available for some of your libraries. ×

LATERINSTALL MANUALLYINSTALL ALL

ℹ Updates are available for some of your boards. ×

LATERINSTALL MANUALLYINSTALL ALL

OfflineLn 1, Col 1LilyGo T-Display on COM10🔔3🏠

TaskStatus | Arduino IDE 2.1.0

File Edit Sketch Tools Help

New Sketch Ctrl+N
New Cloud Sketch Alt+Ctrl+N
Open... Ctrl+O
Open Recent
Sketchbook
Examples
Close Ctrl+W
Save Ctrl+S
Save As... Ctrl+Shift+S
Preferences... Ctrl+Comma
Advanced
Quit Ctrl+Q

esp32 by Espressif Systems
3.2.0 installed
Boards included in this package:
Pycom GPy, ATD1.47-53, YelloByte YB-ESP32-53-AMP (Rev.3), Heltec Wireles...
More Info
3.2.0 REMOVE

TaskStatus.ino

72
73 void TaskBlink(void *pvParameters) // This is a task.
74 {
75 (void) pvParameters;
76 pinMode(LED_BUILTIN, OUTPUT);
77 for (;;) // A Task shall never return or exit.
78 {
79 //Serial.println(11);
80 digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
81 vTaskDelay(1000 / portTICK_PERIOD_MS); // wait for one second
82 digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW
83 vTaskDelay(1000 / portTICK_PERIOD_MS); // wait for one second
84 }
85
86

Output

Ln 1, Col 1 LilyGo T-Display on COM10 3

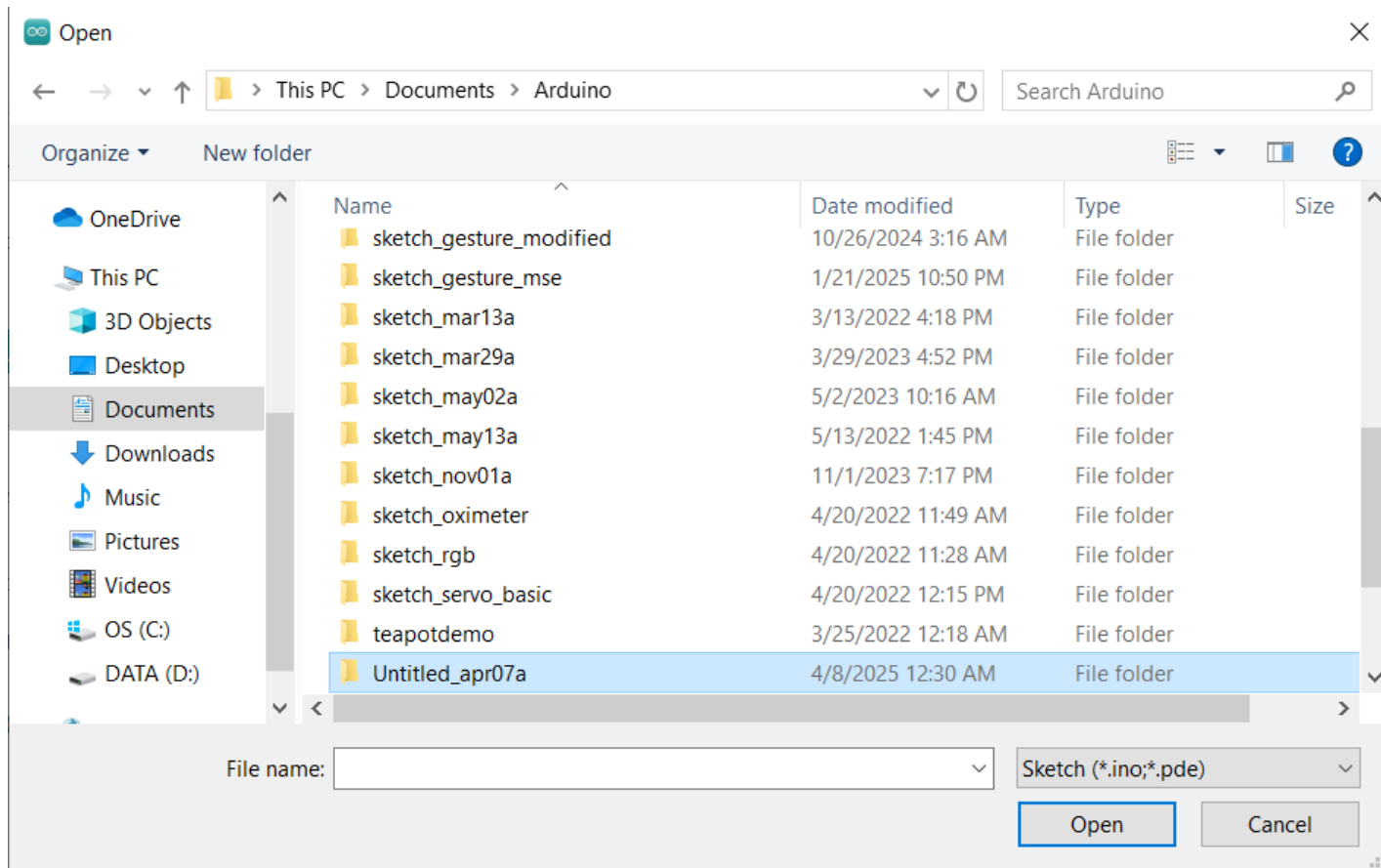
12:37 AM
4/8/2025

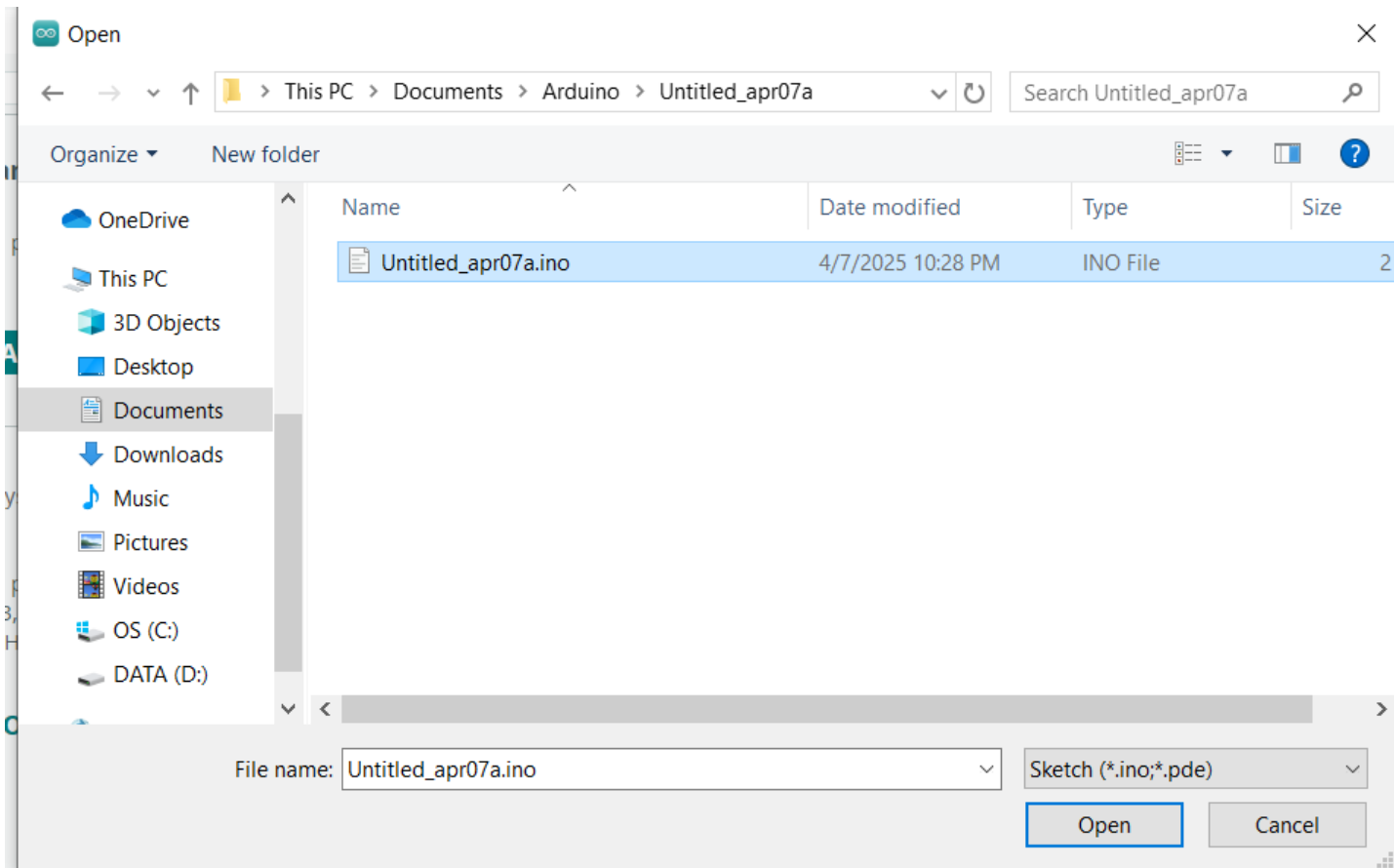
Updates are available for some of your libraries.

LATER INSTALL MANUALLY INSTALL ALL

Updates are available for some of your boards.

LATER INSTALL MANUALLY INSTALL ALL





```
1 #define SECRET_DEVICE_KEY "enkDu2!8EZOCYQeRd7u59u46?"
2 #define SECRET_OPTIONAL_PASS "Your wifi password"
3 #define SECRET_SSID "your wifi name"
4
```

Message (Enter to send message to 'LilyGo T-Display' on 'COM10')

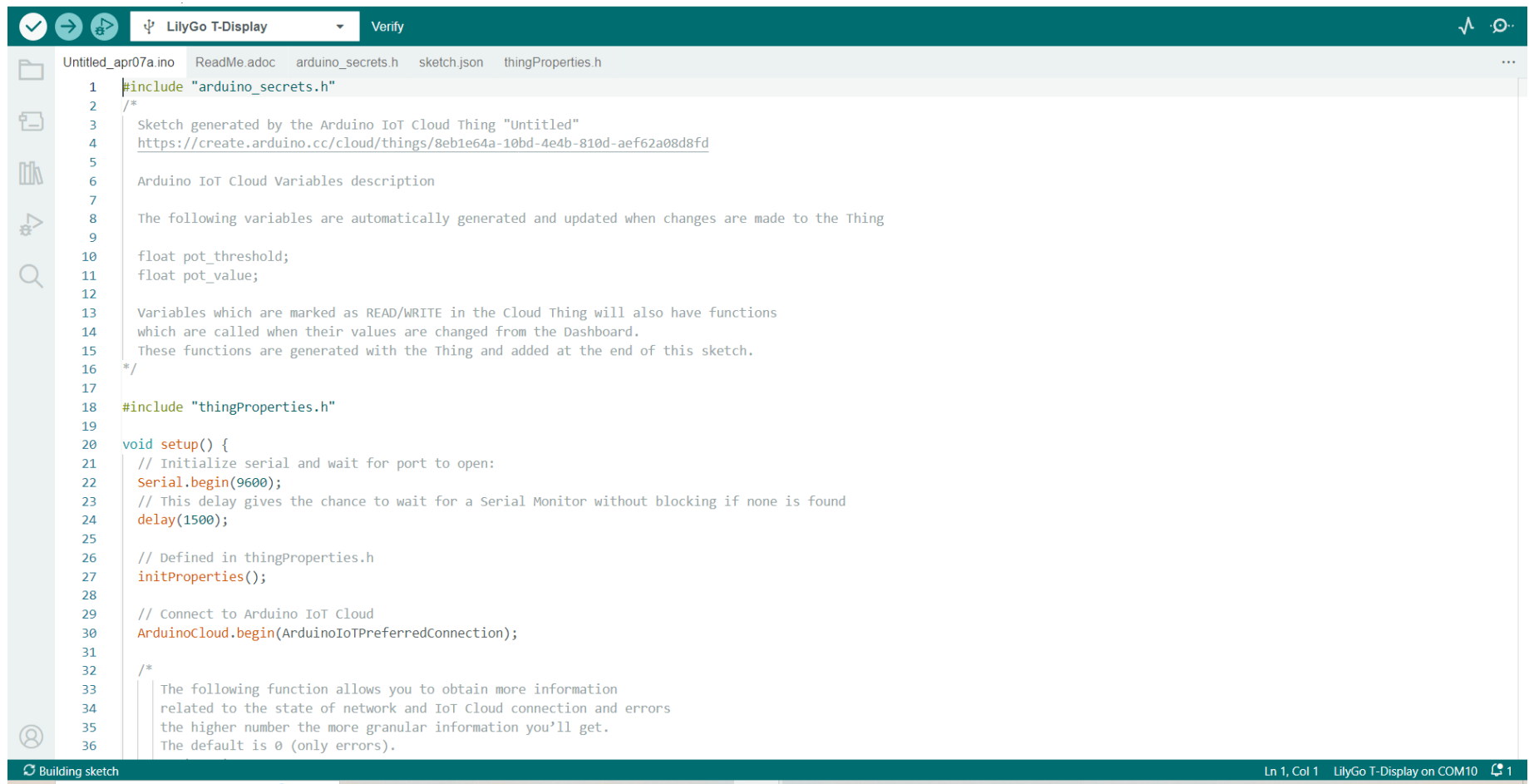
New Line

9600 baud

```

000f000000E (167323) wifi:sta is connecting, return error
Connection to "enkDu2!8EZOCYQeRd7u59u46?" failed
Retrying in "4000" milliseconds
E (171324) wifi:sta is connecting, return error
Connection to "enkDu2!8EZOCYQeRd7u59u46?" failed
Retrying in "4000" milliseconds
E (175325) wifi:sta is connecting, return error
Connection to "enkDu2!8EZOCYQeRd7u59u46?" failed
Retrying in "4000" milliseconds
E (179326) wifi:sta is connecting, return error
Connection to "enkDu2!8EZOCYQeRd7u59u46?" failed

```



66 | 351191.DCE411(2000).
Output Serial Monitor ✕

Message (Enter to send message to 'LilyGo T-Display' on 'COM10')

```
Connection to "enkDu2!8EZOCYQeRd/u59u46?" failed
Retrying in "4000" milliseconds
***** Arduino IoT Cloud - 2.4.1 *****
Device ID: fd085f34-a066-4f50-b348-137e83b289b1
MQTT Broker: iot.arduino.cc:8884
WiFi.status(): 254
Connection to "Orange-7BA3" failed
Retrying in "4000" milliseconds
Connected to "Orange-7BA3"
Connected to Arduino IoT Cloud
Thing ID: 8eb1e64a-10bd-4e4b-810d-aef62a08d8fd
```



←

↺

https://app.arduino.cc/devices

🏠

☆

☆

⋮

CS

csedmicroproce...

csed.microprocessor...

🏠 Home

</> Sketches

📁 Devices

🔔 Things

📊 Dashboards

🔒 ⚙️ Triggers

📄 Templates

📖 Resources

🎓 Courses

🔗 Integrations

⚡ Plan Usage

Arduino Cloud

Ready to go beyond this free

● All Systems Operational

📁 Devices

🔍 Search and filter Devices

🔗 🔧

View 📄

+ DEVICE

☐ Device Name ↑	Status	Type	Associated Thing	Connectivity module
as-test	● ONLINE	ESP32-WROOM-DA Module	as-test	-

1:28 AM